

QUANTITATIVE FINANCE

June 2, 2026

CONTENTS

1. Probability and Measure-Theoretic Foundations	2
1.1. Measure theory	2
1.2. Probability in terms of measure theory	4
1.3. Modes of Convergence	7
1.4. Limit Theorems	8
2. Stochastic Calculus	8
2.1. Brownian Motion and Itô Integration	8
2.2. Itô's Formula and Itô Processes	9
3. Derivatives Pricing Theory	10
3.1. Market Models and No-Arbitrage	10
3.2. Change of Measure	11
3.3. Feynman–Kac and PDE Representation	12
4. Black–Scholes–Merton Model	12
4.1. Model Setup and PDE	12
4.2. Closed-Form Option Prices	13
5. Greeks and Volatility	13
6. Volatility Models	14
6.1. Local Volatility	14
6.2. Stochastic Volatility	15
7. Exotic Options	15
8. Martingales in Finance	16
9. Fixed Income	17

10. Risk Management	18
11. Portfolio Theory	19
12. Numerical Methods for Pricing	19
13. Time Series Models for Finance	21

1. PROBABILITY AND MEASURE-THEORETIC FOUNDATIONS

1.1. Measure theory.

Definition 1.1.1 (Sigma-algebra and measure). Let X be a set. A collection $\mathcal{A}$ of subsets of X is called a **σ -algebra on X** if:

1. $\emptyset \in \mathcal{A}$ and $X \in \mathcal{A}$,
2. if $A \in \mathcal{A}$ then its complement $X \setminus A \in \mathcal{A}$,
3. if $(A_n)_{n \geq 1}$ is a sequence with each $A_n \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

A pair $(X, \mathcal{A})$ is called a **measurable space**.

Definition 1.1.2. Let $(X, \mathcal{A})$ be a measurable space (Definition 1.1.1). A function $\mu : \mathcal{A} \rightarrow [0, \infty]$ is called a **measure on $(X, \mathcal{A})$** if:

1. $\mu(\emptyset) = 0$,
2. for every sequence $(A_n)_{n \geq 1}$ of pairwise disjoint sets in $\mathcal{A}$, one has

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(X, \mathcal{A}, \mu)$ is then called a **measure space**.

Definition 1.1.3 (Probability measure space). A measure space (Definition 1.1.2) $(\Omega, \mathcal{F}, P)$ is called a **probability measure space**, and P is called a **probability measure** if $P(\Omega) = 1$.

In this case, Ω is called the **space of outcomes**, elements of $\mathcal{F}$ are called **events** and P is called the **probability measure**.

Definition 1.1.4 (Borel σ -algebra). Let $(X, \mathcal{T})$ be a topological space. A σ -algebra (Definition 1.1.1) $\mathcal{B}$ on X is called the **Borel σ -algebra (associated with $\mathcal{T}$)** if:

1. $\mathcal{T} \subseteq \mathcal{B}$, i.e. every open set in X is an element of $\mathcal{B}$,
2. $\mathcal{B}$ is contained in every σ -algebra on X that contains $\mathcal{T}$ (i.e. it is the smallest σ -algebra on X containing all open sets).

Equivalently, the Borel σ -algebra on X is the intersection of all σ -algebras on X that contain the topology $\mathcal{T}$. We often denote the Borel σ -algebra on X by

$$\mathcal{B}(X)$$

or, when the topology $\mathcal{T}$ is emphasized or there is potential ambiguity, by

$$\mathcal{B}(X, \mathcal{T}).$$

A *Borel set of $(X, \mathcal{T})$* refers to an element of $\mathcal{B}(X)$.

(♠ TODO: topology on extended real line) In particular, one speaks of the *Borel σ -algebras of $\mathbb{R}$, $[-\infty, \infty]$, and $\mathbb{C}$* where each of these sets are equipped with their usual topological space structures.

Definition 1.1.5. Let $(X, \mathcal{A})$ be a measurable space (Definition 1.1.1).

1. Let $(Y, \mathcal{B})$ be a measurable space. A function $f : X \rightarrow Y$ is said to be *$(\mathcal{A}, \mathcal{B})$ -measurable*, *measurable with respect to the σ -algebras $\mathcal{A}$ and $\mathcal{B}$* , or simply *measurable* if $f^{-1}(B) \in \mathcal{A}$ for all $B \in \mathcal{B}$.
2. A function $f : X \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ is called a *measurable function into the extended real line* if it is a measurable function with respect to $\mathcal{A}$ and the Borel σ -algebra (Definition 1.1.4) of $\overline{\mathbb{R}}$. Explicitly, f is measurable if and only if $f^{-1}((a, \infty]) \in \mathcal{A}$ for every $a \in \mathbb{R}$.
3. A function $f : X \rightarrow \mathbb{C}$ is called a *measurable function into the complex plane* if it is a measurable function with respect to $\mathcal{A}$ and the Borel σ -algebra of $\mathbb{C}$. Equivalently, f is measurable if both $\operatorname{Re}(f)$ and $\operatorname{Im}(f)$ are measurable real-valued functions.

Definition 1.1.6 (Measurable function and Lebesgue integral). (♠ TODO: more genral measurable functions) Let $(X, \mathcal{A}, \mu)$ be a measure space (Definition 1.1.1) and let $f : X \rightarrow \mathbb{R} \cup \{-\infty\} \cup \{\infty\}$ be a function.

1. The function f is called *measurable* if $f^{-1}((-\infty, t)) \in \mathcal{A}$ for every real number t .
2. A *simple function on X* is a measurable function $s : X \rightarrow [0, \infty)$ that takes only finitely many values, i.e. $s = \sum_{k=1}^m a_k \mathbf{1}_{A_k}$ with $a_k \in [0, \infty)$ and $A_k \in \mathcal{A}$ pairwise disjoint.

For such s , the *Lebesgue integral of s with respect to μ* is defined by

$$\int_X s \, d\mu := \sum_{k=1}^m a_k \mu(A_k).$$

(♠ TODO: interval notation for including ∞)

3. For a general nonnegative measurable function $f : X \rightarrow [0, \infty]$, the *Lebesgue integral of f* is defined by

$$\int_X f \, d\mu := \sup \left\{ \int_X s \, d\mu : s \text{ is a simple function with } 0 \leq s \leq f \right\}.$$

4. A measurable function $f : X \rightarrow [-\infty, \infty]$ is said to be *(Lebesgue) integrable* or *(Lebesgue) μ -integrable* if $\int_X |f| \, d\mu < \infty$. In this case, one defines its *Lebesgue integral* by

$$\int_X f \, d\mu := \int_X f^+ \, d\mu - \int_X f^- \, d\mu,$$

where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.

Definition 1.1.7. Let $(X, \mathcal{A}, \mu)$ be a measure space.

1. Let $1 \leq p < \infty$. Given a measurable function $f : X \rightarrow \mathbb{C}$, the **L^p -norm** is the norm defined by

$$\|f\|_p = \left(\int_X |f(x)|^p d\mu(x) \right)^{\frac{1}{p}}.$$

2. Let $p = \infty$. Given a measurable function $f : X \rightarrow \mathbb{C}$, the **L^∞ -norm** is the norm defined by

$$\|f\|_\infty = \inf\{M \geq 0 : |f(x)| \leq M \text{ for } \mu\text{-almost every } x \in X\}.$$

3. Let $1 \leq p \leq \infty$. The **L^p space on X** is defined as the set

$$L^p(X, \mathcal{A}, \mu) = L^p(X) := L^p(X, \mu) = \{f : X \rightarrow \mathbb{C} \mid f \text{ is measurable and } \|f\|_p < \infty\}.$$

An element $f \in L^p(X)$ is called an **L^p -function on the measure space X** .

1.2. Probability in terms of measure theory.

Definition 1.2.1 (Random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3).

A function $X : \Omega \rightarrow S$ into a measurable space (Definition 1.1.1) $(S, \mathcal{S})$ is called a **$(S, \mathcal{S})$ -valued random variable on $(\Omega, \mathcal{F}, P)$** if it is a measurable function (Definition 1.1.5) from $(\Omega, \mathcal{F})$ into $(S, \mathcal{S})$. That is, $X^{-1}(A) \in \mathcal{F}$ for every $A \in \mathcal{S}$.

Special cases:

1. A function $X : \Omega \rightarrow \mathbb{R}$ is called a **real-valued random variable on $(\Omega, \mathcal{F}, P)$** if it is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ -valued, where $\mathcal{B}(\mathbb{R})$ is the Borel σ -algebra on $\mathbb{R}$ (Definition 1.1.4).
2. A function $X : \Omega \rightarrow \mathbb{C}$ is called a **complex-valued random variable on $(\Omega, \mathcal{F}, P)$** if it is $(\mathbb{C}, \mathcal{B}(\mathbb{C}))$ -valued, where $\mathcal{B}(\mathbb{C})$ is the Borel σ -algebra on $\mathbb{C}$ (Definition 1.1.4).
3. A function $X : \Omega \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}$ is called a **random variable taking values in the extended real line on $(\Omega, \mathcal{F}, P)$** if it is $(\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$ -valued, where $\mathcal{B}(\overline{\mathbb{R}})$ is the Borel σ -algebra on $\overline{\mathbb{R}}$.

A function X as above ($(S, \mathcal{S})$ -valued, real-valued, complex-valued, or extended real-valued) is called simply a **random variable on $(\Omega, \mathcal{F}, P)$** when the codomain and σ -algebra are clear from context.

Definition 1.2.2 (Expectation of a random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3).

1. If $X : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable, its **expectation** (or **expected value**) is defined by

$$\mathbb{E}[X] := \int_{\Omega} X dP,$$

where the integral on the right-hand side is the Lebesgue integral (Definition 1.1.6) with respect to the probability measure P .

More explicitly, one defines

- (a) If $X : \Omega \rightarrow [0, \infty)$ is a nonnegative random variable, then

$$\mathbb{E}[X] := \int_{\Omega} X dP,$$

where the integral on the right-hand side is the Lebesgue integral (Definition 1.1.6) with respect to the probability measure P .

- (b) If $X : \Omega \rightarrow \mathbb{R}$ is a real-valued random variable, then

$$\mathbb{E}[X] := \mathbb{E}[X^+] - \mathbb{E}[X^-],$$

where we decompose $X = X^+ - X^-$ by letting $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$; provided that at least one of $\mathbb{E}[X^+]$, $\mathbb{E}[X^-]$ is finite, this is defined as a value in $[-\infty, \infty]$. If both are finite (i.e., X is a P -integrable function (Definition 1.1.6)), then $\mathbb{E}[X]$ is a well-defined real number.

2. More generally, if $(S, \mathcal{S})$ is a measurable space (Definition 1.1.1) and if $X : \Omega \rightarrow S$ is an $(S, \mathcal{S})$ -valued random variable (Definition 1.2.1) with distribution (Definition 1.2.5) μ_X on $(S, \mathcal{S})$, and if $g : S \rightarrow \mathbb{R}$ is a measurable function (Definition 1.1.5) such that $g \in L^1(S, \mathcal{S}, \mu_X)$, then we define the expectation of g :

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) dP(\omega) = \int_S g(x) d\mu_X(x).$$

A random variable X is called **integrable** if $\mathbb{E}[|X|] < \infty$. The space of all integrable real-valued random variables on $(\Omega, \mathcal{F}, P)$ is denoted by $L^1(\Omega, \mathcal{F}, P)$.

Definition 1.2.3 (Variance and covariance). Let $(\Omega, \mathcal{F}, P)$ be a probability space.

1. Let X be an integrable real-valued random variable with $\mathbb{E}[X]$ finite. The **variance of X** is defined by

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

Equivalently, if $\mathbb{E}[X^2] < \infty$, then

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

2. Let X and Y be integrable real-valued random variables with $\mathbb{E}[XY]$ finite. The **covariance of X and Y** is defined by

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

Equivalently, if $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, then

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

The **correlation coefficient of X and Y** (when $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$) is defined by

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}}.$$

A random variable X has zero variance if and only if $X = \mathbb{E}[X]$ almost surely. Covariance is symmetric: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, and it is bilinear in its arguments.

Definition 1.2.4 (Pushforward measure of a measure via a measurable function). (♠ TODO: AI generated, verify) Let $(X, \mathcal{A})$ and $(Y, \mathcal{B})$ be measurable spaces (Definition 1.1.1). Let $f : X \rightarrow Y$ be a measurable function (Definition 1.1.5). Let μ be a measure (Definition 1.1.2) on $(X, \mathcal{A})$.

The *pushforward measure of μ via f* , denoted $f_*\mu$, is the measure on $(Y, \mathcal{B})$ defined by

$$(f_*\mu)(B) = \mu(f^{-1}(B))$$

for every $B \in \mathcal{B}$.

Definition 1.2.5 (Distribution (law) of a random variable on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3), and let $X : \Omega \rightarrow S$ be a $(S, \mathcal{S})$ -valued random variable (Definition 1.2.1) where $(S, \mathcal{S})$ is a measurable space (Definition 1.1.1).

The *distribution* (or *law*) of X , denoted μ_X , is the measure (Definition 1.1.2) on $(S, \mathcal{S})$ defined by

$$\mu_X(A) = P(X^{-1}(A)) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for every $A \in \mathcal{S}$. In other words, μ_X is the pushforward measure (Definition 1.2.4) X_*P of P under X .

In the special case where $S = \mathbb{R}$ and $\mathcal{S} = \mathcal{B}(\mathbb{R})$ is the Borel σ -algebra (Definition 1.1.4) on $\mathbb{R}$, the distribution of X is uniquely determined by its *cumulative distribution function* $F_X : \mathbb{R} \rightarrow [0, 1]$ given by

$$F_X(t) = \mu_X((-\infty, t]) = P(X \leq t).$$

Definition 1.2.6 (Independence of σ -algebras). Let $(\Omega, \mathcal{F}, P)$ be a probability space.

(♠ TODO: sub-sigma algebra) A finite collection of sub- σ -algebras $\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_n$ of $\mathcal{F}$ is called *independent* if for every choice of events $A_i \in \mathcal{G}_i$ ($1 \leq i \leq n$),

$$P\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n P(A_i).$$

An arbitrary (possibly infinite) collection of sub- σ -algebras $\{\mathcal{G}_j\}_{j \in J}$ of $\mathcal{F}$ is called *independent* if every finite subcollection is independent in the sense above. That is, for any distinct indices $j_1, j_2, \dots, j_n \in J$ and any choice of events $A_k \in \mathcal{G}_{j_k}$ ($1 \leq k \leq n$),

$$P\left(\bigcap_{k=1}^n A_k\right) = \prod_{k=1}^n P(A_k).$$

A sequence $(\mathcal{G}_n)_{n=1}^\infty$ of sub- σ -algebras is called an *independent sequence* if the set $\{\mathcal{G}_1, \mathcal{G}_2, \dots\}$ is independent.

Definition 1.2.7 (Independence of random variables on a probability measure space). Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3).

A finite collection of real-valued random variables $X_1, X_2, \dots, X_n$ on $(\Omega, \mathcal{F}, P)$ is called **independent** if the sub- σ -algebras $\sigma(X_1), \sigma(X_2), \dots, \sigma(X_n)$ are independent (Definition 1.2.6), where for each i ,

$$\sigma(X_i) = \{X_i^{-1}(B) : B \in \mathcal{B}(\mathbb{R})\}$$

is the σ -algebra generated by X_i , and $\mathcal{B}(\mathbb{R})$ is the Borel σ -algebra (Definition 1.1.4) on $\mathbb{R}$.

An arbitrary (possibly infinite) collection of random variables $\{X_j\}_{j \in J}$ is called **independent** if every finite subcollection is independent.

A sequence $(X_n)_{n=1}^\infty$ of random variables is called an **independent sequence** if the set $\{\sigma(X_1), \sigma(X_2), \dots\}$ consists of independent σ -algebras.

Equivalently, $X_1, X_2, \dots, X_n$ are independent if and only if for every choice of Borel sets $B_1, B_2, \dots, B_n \in \mathcal{B}(\mathbb{R})$,

$$P\left(\bigcap_{i=1}^n \{\omega : X_i(\omega) \in B_i\}\right) = \prod_{i=1}^n P(\{\omega : X_i(\omega) \in B_i\}).$$

1.3. Modes of Convergence.

Definition 1.3.1 (Almost sure convergence). Let $(X, \mathcal{A}, \mu)$ be a measure space. Let (Y, d_Y) be a metric space (typically $Y = \mathbb{R}, \mathbb{R},$ or $\mathbb{C}$).

Let $(f_n)_{n \geq 1}$ be a sequence of measurable functions (Definition 1.1.5) $f_n : X \rightarrow Y$ and let $f : X \rightarrow Y$ be a function. The sequence (f_n) is said to **converge to f almost surely** or to **converge μ -almost everywhere** if

$$\mu\left(\{x \in X : d_Y(f_n(x), f(x)) \not\rightarrow 0 \text{ as } n \rightarrow \infty\}\right) = 0.$$

Equivalently, $f_n \rightarrow f$ μ -a.e. if there exists a set $N \in \mathcal{A}$ with $\mu(N) = 0$ such that for all $x \in X \setminus N$, $f_n(x) \rightarrow f(x)$ in the metric d_Y .

Definition 1.3.2 (Convergence in distribution). (♠ TODO: cumulative distribution) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3). Let $\{X_n\}_{n=1}^\infty$ and X be real-valued random variables (Definition 1.2.1) on $(\Omega, \mathcal{F}, P)$, with cumulative distribution functions F_{X_n} and F_X , respectively. We say that X_n **converges in distribution** to X (denoted $X_n \xrightarrow{d} X$) if for every $t \in \mathbb{R}$ at which F_X is continuous,

$$\lim_{n \rightarrow \infty} F_{X_n}(t) = F_X(t).$$

Proposition 1.3.3 (Relationships between modes of convergence). Let $(X, \mathcal{A}, \mu)$ be a measure space (Definition 1.1.2), and let $(f_n)_{n=1}^\infty$ and f be measurable functions from X to a metric space (S, d) .

1. If $\mu(X) < \infty$ and $f_n \rightarrow f$ almost surely (i.e., almost everywhere (Definition 1.3.1)), then $f_n \rightarrow f$ almost uniformly.

2. If $f_n \rightarrow f$ almost uniformly, then $f_n \rightarrow f$ in measure.
3. If f_n, f are measurable functions from X to $\mathbb{R}$ and there exists $M > 0$ such that $|f_n(x)| \leq M$ for all n and all $x \in X$, then almost uniform convergence implies convergence in the L^p norm (Definition 1.1.7) for every $1 \leq p < \infty$.
4. If $f_n, f \in L^p(X, \mathcal{A}, \mu)$ for some $1 \leq p < \infty$ and $\|f_n - f\|_p \rightarrow 0$, then there exists a subsequence $(f_{n_k})_{k=1}^\infty$ such that $f_{n_k} \rightarrow f$ almost everywhere (Definition 1.3.1) as $k \rightarrow \infty$.

1.4. Limit Theorems.

Theorem 1.4.1 (Weak law of large numbers (Khinchine)). Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{X_n\}_{n=1}^\infty$ be a sequence of independent and identically distributed real-valued random variables on $(\Omega, \mathcal{F}, P)$, with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2 < \infty$ (Definition 1.2.3).

Let $S_n = X_1 + \cdots + X_n$ and let $\bar{X}_n = S_n/n$. Then for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| \geq \varepsilon) = 0.$$

That is, $\bar{X}_n$ converges to μ in probability. This follows from Chebyshev's inequality applied to $\bar{X}_n$, since $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$.

Theorem 1.4.2 (Strong law of large numbers (Kolmogorov)). Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\{X_n\}_{n=1}^\infty$ be a sequence of independent (Definition 1.2.7) and identically distributed real-valued random variables (Definition 1.2.1) on $(\Omega, \mathcal{F}, P)$ such that $\mathbb{E}[|X_1|] < \infty$.

Let $S_n = \sum_{i=1}^n X_i$ for each $n \geq 1$. Then

$$\frac{S_n}{n} \rightarrow \mathbb{E}[X_1] \quad \text{almost surely as } n \rightarrow \infty.$$

Lemma 1.4.3 (Borel–Cantelli lemmas). Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\{A_n\}_{n=1}^\infty$ be a sequence of events in $\mathcal{F}$. Define the event

$$\limsup_{n \rightarrow \infty} A_n := \bigcap_{n=1}^\infty \bigcup_{k=n}^\infty A_k = \{\omega \in \Omega : \omega \in A_n \text{ for infinitely many } n\}.$$

First lemma If $\sum_{n=1}^\infty P(A_n) < \infty$, then

$$P(\limsup_{n \rightarrow \infty} A_n) = 0.$$

Second lemma If the events A_n are independent and $\sum_{n=1}^\infty P(A_n) = \infty$, then

$$P(\limsup_{n \rightarrow \infty} A_n) = 1.$$

2. STOCHASTIC CALCULUS

2.1. Brownian Motion and Itô Integration.

Definition 2.1.1 (Standard Brownian motion on a probability space). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3). A family

of real-valued random variables $\{B_t\}_{t \geq 0}$ on $(\Omega, \mathcal{F}, P)$ is called a **standard Brownian motion** (*Wiener process*) if it satisfies:

1. $B_0 = 0$ almost surely;
2. (Independent increments) For all $0 \leq t_0 < t_1 < \dots < t_n$, the random variables $B_{t_1} - B_{t_0}, B_{t_2} - B_{t_1}, \dots, B_{t_n} - B_{t_{n-1}}$ are independent;
3. (Gaussian increments) For all $s < t$, the increment $B_t - B_s$ has a normal distribution $\mathcal{N}(0, t - s)$;
4. (Continuous paths) Almost surely, the map $t \mapsto B_t(\omega)$ is continuous on $[0, \infty)$.

For each $t \geq 0$, $\mathbb{E}[B_t] = 0$ and $\text{Var}(B_t) = t$. The process has stationary increments.

Definition 2.1.2 (Itô integral with respect to Brownian motion). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{t \geq 0}$.

For an adapted process $X = \{X_t\}_{t \geq 0}$ such that $\int_0^T \mathbb{E}[X_t^2] dt < \infty$ for all $T > 0$, the **Itô integral of X with respect to B** over $[0, T]$ is denoted

$$\int_0^T X_s dB_s,$$

and is defined as the L^2 -limit of Riemann–Itô sums:

$$\sum_{i=0}^{n-1} X_{t_i} (B_{t_{i+1}} - B_{t_i}),$$

where $0 = t_0 < t_1 < \dots < t_n = T$ and the mesh $\max_i |t_{i+1} - t_i| \rightarrow 0$. The evaluation at the left endpoint X_{t_i} is essential for Itô calculus.

The Itô integral process $M_t = \int_0^t X_s dB_s$ is a continuous martingale with quadratic variation $\langle M \rangle_t = \int_0^t X_s^2 ds$.

Theorem 2.1.3 (Itô isometry for Itô integrals). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{t \geq 0}$. Let $X = \{X_t\}_{t \geq 0}$ be an adapted process with $\int_0^T \mathbb{E}[X_s^2] ds < \infty$ for all $T > 0$.

Then the Itô integral $M_T = \int_0^T X_s dB_s$ satisfies

$$\mathbb{E} \left[\left(\int_0^T X_s dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^T X_s^2 ds \right].$$

More generally, for each $t \geq 0$, the process $M_t^2 - \int_0^t X_s^2 ds$ is a martingale.

2.2. Itô's Formula and Itô Processes.

Theorem 2.2.1 (Itô's formula for a function of an Itô process). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ be a filtered probability space supporting a standard Brownian

motion $\{B_t\}_{t \geq 0}$. Let $X = \{X_t\}_{t \geq 0}$ be an Itô process satisfying the stochastic differential equation

$$dX_t = \mu_t dt + \sigma_t dB_t,$$

where μ_t and σ_t are adapted processes with $\int_0^T (|\mu_s| + \sigma_s^2) ds < \infty$ a.s.

Let $f : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ be of class $C^{1,2}$ (once continuously differentiable in time and twice in space). Then

$$\begin{aligned} df(t, X_t) &= \frac{\partial f}{\partial t}(t, X_t) dt + \frac{\partial f}{\partial x}(t, X_t) dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) (dX_t)^2 \\ &= \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{1}{2} \sigma_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dB_t. \end{aligned}$$

In particular, $(dX_t)^2 = \sigma_t^2 dt$, $dt \cdot dB_t = 0$, and $(dB_t)^2 = dt$.

Definition 2.2.2 (Geometric Brownian motion on a filtered probability space). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{t \geq 0}$. Let $S_0 > 0$, $\mu \in \mathbb{R}$, and $\sigma > 0$.

The process $\{S_t\}_{t \geq 0}$ defined by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

is called a *geometric Brownian motion with drift μ and volatility σ* . By Itô's formula applied to $f(t, x) = \log x$, the explicit solution is

$$S_t = S_0 \exp \left(\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma B_t \right).$$

For each $t > 0$, $\log(S_t/S_0) \sim \mathcal{N}((\mu - \frac{1}{2} \sigma^2)t, \sigma^2 t)$ and

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1).$$

3. DERIVATIVES PRICING THEORY

3.1. Market Models and No-Arbitrage.

Definition 3.1.1 (Market model with risk-free asset and risky assets). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ be a filtered probability space. A *market model on $[0, T]$* consists of:

1. A risk-free asset (bank account) with price process $\{B_t\}_{0 \leq t \leq T}$ satisfying

$$dB_t = r_t B_t dt, \quad B_0 = 1,$$

where r_t is the short rate process. In the constant-rate case, $B_t = e^{rt}$.

2. d risky assets (stocks) with price processes $\{S_t^i\}_{0 \leq t \leq T}$ for $i = 1, \dots, d$, each adapted to $(\mathcal{F}_t)_{t \geq 0}$.

A **trading strategy** is a predictable process $\theta_t = (\theta_t^1, \dots, \theta_t^d)$ specifying the number of shares held in each asset. The associated **self-financing portfolio value** is

$$V_t = \theta_0^B(t)B_t + \sum_{i=1}^d \theta_t^i S_t^i,$$

where $\theta_0^B(t)$ is the number of units in the risk-free asset.

The market is said to be **arbitrage-free** if there exists no self-financing strategy with $V_0 = 0$, $P(V_T \geq 0) = 1$, and $P(V_T > 0) > 0$.

Theorem 3.1.2 (Fundamental theorem of asset pricing for continuous-time models). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ be a filtered probability space. Let $\{S_t^i\}_{0 \leq t \leq T}$ for $i = 1, \dots, d$ be the price processes of risky assets and let B_t be the risk-free asset with constant rate r .

The market is arbitrage-free if and only if there exists an equivalent probability measure $\mathbb{Q} \sim P$ (called an **equivalent martingale measure** or **risk-neutral measure**) such that for each $i = 1, \dots, d$, the discounted asset price process

$$\tilde{S}_t^i = e^{-rt} S_t^i$$

is a $\mathbb{Q}$ -martingale with respect to $(\mathcal{F}_t)_{0 \leq t \leq T}$.

If, in addition, such a measure is unique, the market is called **complete**, and every European contingent claim has a unique arbitrage-free price given by

$$V_0 = \mathbb{E}^{\mathbb{Q}} [e^{-rT} \text{payoff}(S_T)].$$

3.2. Change of Measure.

Theorem 3.2.1 (Girsanov's theorem for change of measure on Brownian motion). (♠ **TODO: AI generated, verify**) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{0 \leq t \leq T}$. Let $\theta = \{\theta_t\}_{0 \leq t \leq T}$ be an adapted process satisfying the Novikov condition:

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \theta_s^2 ds \right) \right] < \infty.$$

Define the stochastic exponential (Doléans-Dade exponential)

$$Z_t = \exp \left(- \int_0^t \theta_s dB_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right), \quad 0 \leq t \leq T.$$

Then Z is a true martingale. Define the measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_T)$ by

$$\left. \frac{d\mathbb{Q}}{dP} \right|_{\mathcal{F}_t} = Z_t.$$

Under $\mathbb{Q}$, the process

$$W_t = B_t + \int_0^t \theta_s ds$$

is a standard Brownian motion (called a $\mathbb{Q}$ -Brownian motion). The drift of any Itô process is shifted by $\sigma\theta$ under this change of measure.

3.3. Feynman–Kac and PDE Representation.

Theorem 3.3.1 (Feynman–Kac formula for parabolic PDEs and Itô processes). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{t \geq 0}$. Let $X = \{X_s\}_{s \in [t, T]}$ satisfy the SDE

$$dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dB_s, \quad X_t = x.$$

Let $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be bounded measurable functions. Consider the parabolic PDE

$$\frac{\partial u}{\partial t} + \mu(t, x) \frac{\partial u}{\partial x} + \frac{1}{2} \sigma^2(t, x) \frac{\partial^2 u}{\partial x^2} - f(t, x)u + g(t, x) = 0$$

with terminal condition $u(T, x) = 0$.

Then the unique bounded solution is given by

$$u(t, x) = \mathbb{E} \left[\int_t^T e^{-\int_t^\tau f(\tau, X_\tau) d\tau} g(s, X_s) ds \mid X_t = x \right].$$

In the special case $f \equiv 0$ and terminal condition $u(T, x) = \Phi(x)$,

$$u(t, x) = \mathbb{E} [\Phi(X_T) \mid X_t = x]$$

solves $\frac{\partial u}{\partial t} + \mu \frac{\partial u}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 u}{\partial x^2} = 0$.

4. BLACK–SCHOLES–MERTON MODEL

4.1. Model Setup and PDE.

Definition 4.1.1 (Black–Scholes–Merton market model). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ be a filtered probability space supporting a standard Brownian motion $\{B_t\}_{t \geq 0}$. The Black–Scholes–Merton model consists of:

1. A risk-free asset with constant rate $r \in \mathbb{R}$, price process $B_t = e^{rt}$.
2. A risky asset (stock) following a geometric Brownian motion (Definition 2.2.2):

$$dS_t = \mu S_t dt + \sigma S_t dB_t, \quad S_0 > 0,$$

where $\mu \in \mathbb{R}$ is the drift and $\sigma > 0$ is the volatility.

The model assumes no transaction costs, continuous trading, no dividends (unless otherwise specified), and that σ and r are constant. Under these assumptions the market is complete and admits a unique equivalent martingale measure.

Theorem 4.1.2 (Black–Scholes–Merton PDE for option prices). (♠ TODO: AI generated, verify) In the Black–Scholes–Merton model (Definition 4.1.1), let $V(t, S)$ denote the price at time t of a derivative with payoff $\Phi(S_T)$ at maturity T , where the underlying stock price is S .

Then $V : [0, T] \times (0, \infty) \rightarrow \mathbb{R}$ satisfies the Black–Scholes–Merton partial differential equation

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0,$$

with terminal condition $V(T, S) = \Phi(S)$.

The unique bounded solution is given by the risk-neutral pricing formula:

$$V(t, S) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}} [\Phi(S_T) \mid S_t = S],$$

where under $\mathbb{Q}$, $dS_u = rS_u du + \sigma S_u dW_u$ and W is a $\mathbb{Q}$ -Brownian motion.

4.2. Closed-Form Option Prices.

Theorem 4.2.1 (Black–Scholes–Merton formula for European call and put options). (♠ TODO: AI generated, verify) In the Black–Scholes–Merton model (Definition 4.1.1), let $K > 0$ be a strike price and $T > 0$ a maturity. Define

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

The arbitrage-free prices of a European call and put option are:

$$C_0 = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2),$$

$$P_0 = K e^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1),$$

where Φ is the cumulative distribution function of the standard normal distribution.

The call–put parity relation holds: $C_0 - P_0 = S_0 - K e^{-rT}$.

5. GREEKS AND VOLATILITY

Definition 5.0.1 (The Greeks: delta, gamma, theta, vega, rho for option prices). (♠ TODO: AI generated, verify) In the Black–Scholes–Merton model (Definition 4.1.1), let $V(t, S; r, \sigma)$ denote the price of a derivative at time t with underlying stock price S , risk-free rate r , and volatility σ .

The sensitivity measures (called the *Greeks*) are defined as:

1. **Delta**: $\Delta = \frac{\partial V}{\partial S}$ — sensitivity to the underlying price.
2. **Gamma**: $\Gamma = \frac{\partial^2 V}{\partial S^2}$ — curvature with respect to the underlying price.
3. **Theta**: $\Theta = \frac{\partial V}{\partial t}$ — sensitivity to the passage of time.
4. **Vega**: $\nu = \frac{\partial V}{\partial \sigma}$ — sensitivity to volatility.
5. **Rho**: $\rho = \frac{\partial V}{\partial r}$ — sensitivity to the risk-free rate.

For a European call option in the Black–Scholes model:

$$\Delta_{\text{call}} = \Phi(d_1), \quad \Gamma_{\text{call}} = \frac{\phi(d_1)}{S\sigma\sqrt{T-t}}, \quad \nu_{\text{call}} = S\sqrt{T-t}\phi(d_1),$$

where $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ is the standard normal density.

A portfolio is *delta-neutral* if its total delta is zero; it is *gamma-neutral* (resp. *vega-neutral*) if its total gamma (resp. vega) is zero.

Definition 5.0.2 (Implied volatility of a traded option). (♠ TODO: AI generated, verify) In the Black–Scholes–Merton model (Definition 4.1.1), let $V_{\text{BS}}(t, S; K, T, r, \sigma)$ denote the Black–Scholes price of a European option with strike K and maturity T .

Given an observed market price V_{mkt} for an option with parameters (S, K, T, r) , the *implied volatility* is the unique value $\sigma_{\text{imp}} > 0$ such that

$$V_{\text{BS}}(t, S; K, T, r, \sigma_{\text{imp}}) = V_{\text{mkt}}.$$

The mapping $\sigma \mapsto V_{\text{BS}}(\sigma)$ is strictly increasing (vega > 0 for non-expired, non-degenerate options), so the implied volatility exists and is unique.

The *volatility smile* refers to the empirical observation that σ_{imp} plotted against strike K typically exhibits a U-shaped curve, contradicting the constant-volatility assumption of Black–Scholes. The *volatility surface* is the function $(K, T) \mapsto \sigma_{\text{imp}}(K, T)$.

6. VOLATILITY MODELS

6.1. Local Volatility.

Definition 6.1.1 (Dupire’s local volatility model for option pricing). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ be a filtered probability space under the risk-neutral measure. Let $S = \{S_t\}_{t \geq 0}$ follow the SDE

$$dS_t = rS_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t,$$

where W is a $\mathbb{Q}$ -Brownian motion and $\sigma_{\text{loc}} : [0, \infty) \times (0, \infty) \rightarrow (0, \infty)$ is the *local volatility surface*.

Given an arbitrage-free implied volatility surface $\sigma_{\text{imp}}(K, T)$, Dupire’s formula determines the local volatility:

$$\sigma_{\text{loc}}^2(K, T) = \frac{\frac{\partial V}{\partial T} + rK \frac{\partial V}{\partial K}}{\frac{1}{2}K^2 \frac{\partial^2 V}{\partial K^2}}$$

where $V(K, T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ is the European call price. The partial derivatives are evaluated at strike K and maturity T .

The local volatility model exactly reproduces the observed market prices of all vanilla options, but typically generates a volatility smile that differs from the observed dynamics.

6.2. Stochastic Volatility.

Definition 6.2.1 (Heston stochastic volatility model for asset prices). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ be a filtered probability space under the risk-neutral measure supporting two correlated Brownian motions W^S and W^v with $d\langle W^S, W^v \rangle_t = \rho dt$, where $\rho \in [-1, 1]$.

The Heston model specifies the stock price process S and variance process v by the coupled SDEs:

$$dS_t = rS_t dt + \sqrt{v_t}S_t dW_t^S,$$

$$dv_t = \kappa(\theta - v_t) dt + \xi\sqrt{v_t} dW_t^v,$$

where:

1. $\kappa > 0$ is the mean reversion speed of variance;
2. $\theta > 0$ is the long-run mean level of variance;
3. $\xi > 0$ is the volatility of variance (vol-of-vol);
4. $v_0 > 0$ is the initial variance.

The Feller condition $2\kappa\theta \geq \xi^2$ ensures that v_t remains strictly positive. When this condition fails, v_t may hit zero, in which case the behavior at zero (reflecting or absorbing) must be specified.

The model admits a semi-closed-form characteristic function for $\log S_T$, enabling option prices to be computed via Fourier inversion (Carr–Madan formula).

7. EXOTIC OPTIONS

Definition 7.0.1 (Barrier options and lookback options). (♠ TODO: AI generated, verify) In the Black–Scholes–Merton model (Definition 4.1.1), let $S = \{S_t\}_{0 \leq t \leq T}$ be the stock price process.

A *barrier option* is a path-dependent derivative whose payoff depends on whether the underlying asset price crosses a predetermined barrier level $H > 0$ during $[0, T]$. The main types are:

1. *Up-and-out call*: pays $(S_T - K)^+$ if $\max_{0 \leq t \leq T} S_t < H$ and $H > S_0$; zero otherwise.
2. *Down-and-out put*: pays $(K - S_T)^+$ if $\min_{0 \leq t \leq T} S_t > H$ and $H < S_0$; zero otherwise.
3. *Up-and-in call*: pays $(S_T - K)^+$ if $\max_{0 \leq t \leq T} S_t \geq H$ and $H > S_0$; zero otherwise.
4. *Down-and-in put*: pays $(K - S_T)^+$ if $\min_{0 \leq t \leq T} S_t \leq H$ and $H < S_0$; zero otherwise.

A *floating lookback call option* has payoff $\max_{0 \leq t \leq T} S_t - S_T$. A *floating lookback put option* has payoff $S_T - \min_{0 \leq t \leq T} S_t$.

8. MARTINGALES IN FINANCE

Definition 8.0.1 (Filtration of σ -algebras on a probability space). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, P)$ be a probability measure space (Definition 1.1.3). Let I be a linearly ordered index set (typically $I = \mathbb{N}_{\geq 0}$ or $I = [0, \infty)$).

A *filtration of σ -algebras on $(\Omega, \mathcal{F}, P)$ indexed by I* is a family $(\mathcal{F}_t)_{t \in I}$ of sub- σ -algebras of $\mathcal{F}$ such that for all $s, t \in I$ with $s \leq t$,

$$\mathcal{F}_s \subseteq \mathcal{F}_t.$$

The pair $(\Omega, (\mathcal{F}_t)_{t \in I}, P)$ is called a *filtered probability space*. The σ -algebra $\mathcal{F}_t$ represents the information available up to time t .

Definition 8.0.2 (Martingale, submartingale, supermartingale with respect to a filtration). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_t)_{t \in I}, P)$ be a filtered probability space (Definition 8.0.1), where I is a linearly ordered index set. Let $\{X_t\}_{t \in I}$ be a family of real-valued random variables such that for each $t \in I$:

1. X_t is $\mathcal{F}_t$ -measurable (i.e., X_t is *adapted to the filtration* $(\mathcal{F}_t)_{t \in I}$).
2. $\mathbb{E}[|X_t|] < \infty$.

The family $\{X_t\}_{t \in I}$ is called:

1. A *martingale with respect to $(\mathcal{F}_t)_{t \in I}$* if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] = X_s \quad \text{a.s.}$$

2. A *submartingale with respect to $(\mathcal{F}_t)_{t \in I}$* if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] \geq X_s \quad \text{a.s.}$$

3. A *supermartingale with respect to $(\mathcal{F}_t)_{t \in I}$* if for all $s \leq t$ in I ,

$$\mathbb{E}[X_t | \mathcal{F}_s] \leq X_s \quad \text{a.s.}$$

Here $\mathbb{E}[\cdot | \mathcal{F}_s]$ denotes the conditional expectation.

Theorem 8.0.3 (Optional stopping theorem for martingales). (♠ TODO: AI generated, verify) Let $(\Omega, (\mathcal{F}_n)_{n \geq 0}, P)$ be a filtered probability space (Definition 8.0.1). Let $\{X_n\}_{n \geq 0}$ be a martingale (Definition 8.0.2) with respect to $(\mathcal{F}_n)_{n \geq 0}$.

Let $T : \Omega \rightarrow \mathbb{N}_{\geq 0} \cup \{\infty\}$ be a stopping time with respect to $(\mathcal{F}_n)_{n \geq 0}$, i.e., $\{T = n\} \in \mathcal{F}_n$ for all $n \geq 0$. If one of the following conditions holds:

1. T is bounded almost surely (i.e., there exists $N < \infty$ such that $T \leq N$ a.s.);
2. $\mathbb{E}[T] < \infty$ and $|X_{n+1} - X_n| \leq M$ a.s. for some constant $M > 0$ and all n ;
3. $\mathbb{E}[|X_T|] < \infty$ and $\lim_{N \rightarrow \infty} \mathbb{E}[|X_N| \mathbf{1}_{\{T > N\}}] = 0$,

then the stopped random variable X_T is integrable and

$$\mathbb{E}[X_T] = \mathbb{E}[X_0].$$

9. FIXED INCOME

Definition 9.0.1 (Zero-coupon bond, yield curve and forward rates). (♠ TODO: AI generated, verify) A *zero-coupon bond maturing at time T* has price $P(t, T)$ at time $t \leq T$, paying 1 unit of currency at maturity. The price satisfies $P(T, T) = 1$ and $0 < P(t, T) \leq 1$.

The *continuously compounded spot rate* (or yield) for maturity T is

$$y(t, T) = -\frac{\log P(t, T)}{T - t}.$$

The map $T \mapsto y(t, T)$ is called the *yield curve*.

The *instantaneous forward rate* for maturity T observed at time t is

$$f(t, T) = -\frac{\partial}{\partial T} \log P(t, T).$$

The spot rate and forward rate are related by

$$y(t, T) = \frac{1}{T - t} \int_t^T f(t, u) du.$$

The *instantaneous short rate* is $r_t = f(t, t) = -\frac{\partial}{\partial T} \log P(t, T)|_{T=t}$.

Definition 9.0.2 (Duration and convexity of a bond). (♠ TODO: AI generated, verify) Let $P(y)$ denote the price of a bond as a function of its yield to maturity y .

The *Macaulay duration* of the bond is

$$D_{\text{Mac}} = -\frac{1}{P} \sum_{i=1}^n t_i \cdot \text{PV}(CF_i),$$

where $\text{PV}(CF_i)$ is the present value of the i -th cash flow occurring at time t_i .

The *modified duration* is

$$D_{\text{mod}} = -\frac{1}{P} \frac{dP}{dy} = \frac{D_{\text{Mac}}}{1 + y/k},$$

where k is the compounding frequency. Modified duration measures the first-order percentage price change per unit change in yield:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \cdot \Delta y.$$

The *convexity* of the bond is

$$C = \frac{1}{P} \frac{d^2 P}{dy^2}.$$

The second-order approximation to the price change is

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \cdot \Delta y + \frac{1}{2} C \cdot (\Delta y)^2.$$

For a zero-coupon bond maturing at time T , Macaulay duration equals T .

Definition 9.0.3 (Vasicek and CIR short-rate models for interest rates). (♠ TODO: AI generated, verify) Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{Q})$ be a filtered probability space under the risk-neutral measure supporting a Brownian motion W .

The *Vasicek model* (also called the Ornstein–Uhlenbeck short-rate model) specifies the short rate by

$$dr_t = a(b - r_t) dt + \sigma dW_t,$$

where $a > 0$ is the speed of mean reversion, $b \in \mathbb{R}$ is the long-run mean, and $\sigma > 0$ is the volatility. The short rate can become negative under this model.

The *CIR model* (Cox–Ingersoll–Ross) specifies

$$dr_t = a(b - r_t) dt + \sigma \sqrt{r_t} dW_t,$$

where the square-root volatility ensures $r_t \geq 0$ when the Feller condition $2ab \geq \sigma^2$ holds.

In both models, zero-coupon bond prices have the affine form $P(t, T) = A(t, T)e^{-B(t, T)r_t}$ with explicit functions A and B .

10. RISK MANAGEMENT

Definition 10.0.1 (Value at Risk (VaR) for a portfolio loss distribution). (♠ TODO: AI generated, verify) Let L be the random variable representing the loss in portfolio value over a fixed time horizon. Let $\alpha \in (0, 1)$ be a confidence level.

The *Value at Risk at level α* , denoted $\text{VaR}_\alpha(L)$, is the α -quantile of the loss distribution:

$$\text{VaR}_\alpha(L) = \inf\{x \in \mathbb{R} : P(L \leq x) \geq \alpha\}.$$

Equivalently, $P(L > \text{VaR}_\alpha(L)) \leq 1 - \alpha$. VaR measures the maximum loss that will not be exceeded with probability α over the given horizon.

VaR does not satisfy the axiom of subadditivity and hence is not a coherent risk measure. Specifically, there exist portfolios L_1, L_2 such that

$$\text{VaR}_\alpha(L_1 + L_2) > \text{VaR}_\alpha(L_1) + \text{VaR}_\alpha(L_2).$$

Definition 10.0.2 (Expected Shortfall (Conditional VaR) for a portfolio loss). (♠ TODO: AI generated, verify) Let L be the random variable representing the loss in portfolio value over a fixed time horizon. Let $\alpha \in (0, 1)$ be a confidence level.

The *Expected Shortfall at level α* , denoted $\text{ES}_\alpha(L)$ (also called *Conditional Value at Risk*), is defined as

$$\text{ES}_\alpha(L) = \mathbb{E}[L \mid L > \text{VaR}_\alpha(L)] = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(L) du,$$

where $\text{VaR}_u(L)$ is the Value at Risk (Definition 10.0.1).

Expected Shortfall measures the average loss in the worst $1 - \alpha$ fraction of cases. Unlike VaR, ES satisfies all four axioms of a coherent risk measure: monotonicity, subadditivity, positive homogeneity, and translation invariance.

11. PORTFOLIO THEORY

Definition 11.0.1 (Markowitz mean-variance portfolio optimization). (♠ TODO: AI generated, verify) Let $R = (R_1, \dots, R_d)^\top$ be the vector of excess returns on d risky assets, with expected return $\mu = \mathbb{E}[R]$ and covariance matrix Σ . Let $w \in \mathbb{R}^d$ be the portfolio weight vector.

The portfolio return is $R_p = w^\top R$, with

$$\mathbb{E}[R_p] = w^\top \mu, \quad \text{Var}(R_p) = w^\top \Sigma w.$$

The *Markowitz mean-variance optimization problem* minimizes portfolio variance for a given target return μ^* :

$$\min_{w \in \mathbb{R}^d} w^\top \Sigma w \quad \text{subject to } w^\top \mu = \mu^*, \quad \sum_{i=1}^d w_i = 1.$$

The set of solutions forms the *efficient frontier* in $(\sigma_p, \mathbb{E}[R_p])$ -space. The *minimum variance portfolio* has weights

$$w_{\text{mv}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}},$$

where $\mathbf{1}$ is the vector of ones.

When a risk-free asset with rate r_f exists, the *tangency portfolio* (market portfolio) has weights

$$w_{\text{tan}} = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{\mathbf{1}^\top \Sigma^{-1}(\mu - r_f \mathbf{1})},$$

and the capital market line gives the optimal risk-return tradeoff.

12. NUMERICAL METHODS FOR PRICING

Definition 12.0.1 (Binomial option pricing tree (Cox–Ross–Rubinstein)). (♠ TODO: AI generated, verify) Let $S_0 > 0$, $r \in \mathbb{R}$, $\sigma > 0$, and $T > 0$. Divide $[0, T]$ into n equal time steps of length $\Delta t = T/n$. The Cox–Ross–Rubinstein (CRR) binomial tree models the stock price as follows:

At each node (i, j) where $i \in \{0, \dots, n\}$ is the time step and $j \in \{0, \dots, i\}$ is the number of up moves, the stock price is

$$S_{i,j} = S_0 u^j d^{i-j},$$

where

$$u = e^{\sigma \sqrt{\Delta t}}, \quad d = e^{-\sigma \sqrt{\Delta t}} = \frac{1}{u}.$$

The risk-neutral up probability is

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

The arbitrage-free price of a European option with payoff $\Phi(S_T)$ at maturity is computed by backward induction:

$$V_{i,j} = e^{-r\Delta t} (pV_{i+1,j+1} + (1-p)V_{i+1,j}), \quad V_{n,j} = \Phi(S_{n,j}).$$

As $n \rightarrow \infty$, the binomial price converges to the Black–Scholes price.

Definition 12.0.2 (Monte Carlo simulation for option pricing with variance reduction). (♠ **TODO: AI generated, verify**) In the Black–Scholes–Merton model (Definition 4.1.1), the price of a European option with payoff $\Phi(S_T)$ is

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}} [\Phi(S_T)].$$

The *Monte Carlo estimator* using N independent simulated paths is

$$\hat{V}_N = \frac{e^{-rT}}{N} \sum_{i=1}^N \Phi(S_T^{(i)}),$$

where each path is generated from the explicit solution under $\mathbb{Q}$:

$$S_T^{(i)} = S_0 \exp \left(\left(r - \frac{1}{2}\sigma^2 \right) T + \sigma\sqrt{T}Z_i \right), \quad Z_i \sim \mathcal{N}(0, 1).$$

By the central limit theorem, $\hat{V}_N$ is asymptotically normal with standard error $\text{StdDev}(\Phi(S_T))e^{-rT}/\sqrt{N}$.

Common *variance reduction techniques*:

1. *Antithetic variates*: pair each Z_i with $-Z_i$ to reduce variance via negative correlation.
2. *Control variates*: use a correlated variable Y with known expectation: $\hat{V}_{\text{CV}} = \hat{V}_N - \beta(Y - \mathbb{E}[Y])$.
3. *Importance sampling*: sample from a tilted distribution to reduce the variance of rare events.

Monte Carlo is particularly effective for high-dimensional path-dependent options where finite difference methods suffer from the curse of dimensionality.

Definition 12.0.3 (Finite difference methods for the Black–Scholes PDE). (♠ **TODO: AI generated, verify**) The Black–Scholes PDE (Theorem 4.1.2) is a parabolic PDE. On a truncated domain $S \in [0, S_{\max}]$ and time grid $t_i = i\Delta t$, the PDE can be discretized using finite differences:

Let $V_j^i \approx V(t_i, S_j)$ where $S_j = j\Delta S$. The *explicit Euler method* gives

$$\frac{V_j^{i+1} - V_j^i}{\Delta t} + rS_j \frac{V_{j+1}^i - V_{j-1}^i}{2\Delta S} + \frac{1}{2}\sigma^2 S_j^2 \frac{V_{j+1}^i - 2V_j^i + V_{j-1}^i}{(\Delta S)^2} - rV_j^i = 0.$$

The method is conditionally stable. The *implicit Euler method* evaluates spatial derivatives at time t_{i+1} , yielding an unconditionally stable tridiagonal system solved by the Thomas algorithm (TDMA). The *Crank-Nicolson scheme* averages explicit and implicit steps for second-order accuracy in both space and time, but may produce spurious oscillations near discontinuous payoffs.

Boundary conditions: $V(t, 0) = 0$ for calls (Ke^{-rt} for puts); $V(t, S_{\max}) = S_{\max} - Ke^{-r(T-t)}$ for calls (0 for deep-in-the-money puts).

13. TIME SERIES MODELS FOR FINANCE

Definition 13.0.1 (ARIMA models for time series data). (♠ TODO: AI generated, verify) Let $\{X_t\}_{t \in \mathbb{Z}}$ be a discrete-time stochastic process.

An *autoregressive process of order p* , denoted $\text{AR}(p)$, satisfies

$$X_t = c + \sum_{i=1}^p \phi_i X_{t-i} + \varepsilon_t,$$

where $\{\varepsilon_t\}$ is white noise with mean zero and variance σ_ε^2 . The process is stationary when the roots of $1 - \sum_{i=1}^p \phi_i z^i = 0$ lie outside the unit circle.

A *moving average process of order q* , denoted $\text{MA}(q)$, satisfies

$$X_t = \mu + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}.$$

An *ARIMA(p, d, q) process* is a process whose d -th differences $\nabla^d X_t = (1 - L)^d X_t$ follow an $\text{ARMA}(p, q)$ model, where L is the lag operator ($LX_t = X_{t-1}$). An $\text{ARIMA}(p, 1, q)$ model is commonly used for non-stationary financial time series such as log prices.

Two assets X_t and Y_t are *cointegrated* if each is integrated of order 1 (i.e., $\text{ARIMA}(\cdot, 1, \cdot)$) but a linear combination $\beta X_t - Y_t$ is stationary ($\text{ARIMA}(\cdot, 0, \cdot)$).

Definition 13.0.2 (GARCH models for conditional heteroskedasticity in returns). (♠ TODO: AI generated, verify) Let $\{r_t\}_{t \in \mathbb{Z}}$ be a sequence of financial returns. Let $(\mathcal{F}_t)_{t \in \mathbb{Z}}$ be the natural filtration.

A *GARCH(p, q) model* (generalized autoregressive conditional heteroskedasticity) specifies

$$r_t = \mu_t + \sigma_t \varepsilon_t,$$

where $\varepsilon_t \sim \mathcal{N}(0, 1)$ i.i.d., and the conditional variance evolves as

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_i r_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2,$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_j \geq 0$, and $\sum_{i=1}^q \alpha_i + \sum_{j=1}^p \beta_j < 1$ for stationarity.

The most widely used specification is GARCH(1, 1):

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2.$$

GARCH models capture volatility clustering: large (small) returns tend to be followed by large (small) returns. The unconditional variance is $\mathbb{E}[\sigma_t^2] = \omega/(1 - \alpha - \beta)$ in the GARCH(1, 1) case.